AF3625 Group 5 — speaker notes

Questions 5.1–5.4 · 26 slides · 16 min 20 s of scripted speech (≈ 20–22 minutes with the diagrams and pauses) · one block per slide, in deck order.

Role	Slides	Time
Presenter A	1–4 · cover, roadmap, concept primitives	2:35
Presenter B	5–11 · Question 5.1 — television market, country Z	4:35
Presenter C	12–19 · Question 5.2 — handbag market, country A	4:50
Presenter D	20–26 · Question 5.3 Egypt + Question 5.4 article	4:25

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1 Title

0:20 · PRESENTER A

Good morning. We are Group 5, and this is our tutorial presentation for AF3625. Over the next twenty minutes we will work through Questions 5.1 to 5.4. All four of them are about the same thing: what happens to a small open economy when its government restricts trade.

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2 Roadmap

0:35 · PRESENTER A

Here is the plan. We start with two minutes of theory that everything else uses — the world price, and the difference between a tariff and an import quota. Then [B] takes the television market in country Z: a diagram question with an import quota. [C] does the handbag algebra, where the government has to choose a tariff. [D] takes Egypt's generator market, and then a real 2024 trade case — the European Union's tariffs on Chinese electric cars. We will take questions at the end.

> Handover: Handover to [B] after the two primer slides.

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3 Primer – world price

0:45 · PRESENTER A

One idea first, because every question depends on it. Country Z and country A are small countries, which means price takers: nothing they buy or sell can move the world price, so the world price is drawn as a flat line. Once they trade, domestic buyers and sellers all face that price. The rule is simple. If the price you would have without trade — the autarky price — is above the world price, you import, because foreign producers can undercut your own. If it is below, you export. And imports are just the gap between what domestic consumers want and what domestic producers will supply at the world price.

4 Primer – surplus and barriers

0:50 · PRESENTER A

Second idea: surplus. Consumer surplus is the area under demand and above the price — the benefit consumers get over what they actually pay. Producer surplus is the area above supply and below the price. Add them and you have social surplus: the total gain the market creates. Now the two barriers we use today. A tariff is a tax on each imported unit: the domestic price becomes the world price plus the tariff, and the government collects that price gap as revenue. A quota caps the quantity: the domestic price rises until the market clears with only that many imports, and the price gap goes to whoever holds the import licences — that is quota rent. Both cost the country the same two deadweight-loss triangles.

> Handover: Handover to [B] — the Q5.1 presenter takes over here.

5 Part 2 · Question 5.1

0:35 · PRESENTER B

Part two is Question 5.1, and I will read it as it is set. Country Z is a small country where all buyers and sellers are price takers; the figure shows the domestic demand and supply curves of the television market; and the country is open for international trade. Six parts: why it imports, how many it imports, producer surplus, the effect of a 20,000-unit quota, the S-plus-quota curve — which the question says is plotted wrongly — and finally quota rent.

6 Q5.1 – the market

0:35 · PRESENTER B

First, read the two lines off the figure. Demand is P equals 225 minus 1.25 Q; supply is P equals 50 plus 1.25 Q, with Q in thousands of televisions. Where they cross is autarky: seventy thousand televisions at a hundred and thirty-seven dollars fifty. The world price is seventy-five dollars, drawn as a horizontal line because country Z cannot influence it. Notice the world price sits below the autarky price — that single fact is the reason this country imports.

7 Q5.1 (i) (ii)

0:35 · PRESENTER B

Part (i) asks why. At seventy-five dollars, domestic producers will supply only twenty thousand televisions, but consumers want a hundred and twenty thousand. Domestic supply on its own cannot satisfy that demand at that price, and foreign producers are willing to sell at seventy-five, so the shortfall is imported. Part (ii): put seventy-five into the supply equation and you get twenty thousand; into demand and you get a hundred and twenty thousand. Imports are the difference — one hundred thousand televisions a year.

8 Q5.1 (iii) producer surplus

0:35 · PRESENTER B

Part (iii) is producer surplus under free trade. It is the triangle above the domestic supply curve and below seventy-five dollars, up to the twenty thousand that domestic producers sell. Base is twenty thousand; height is seventy-five minus fifty, so twenty-five dollars. A half times base times height gives two hundred and fifty thousand dollars. The trap here is using a hundred and twenty thousand as the base — that is what consumers buy, but the surplus belongs to domestic producers only.

9 Q5.1 (iv) quota effect

0:40 · PRESENTER B

Part (iv): the government imposes an import quota of twenty thousand televisions. The intended effect comes in three steps. It caps imports at twenty thousand, far below the hundred thousand that free trade would bring in, so the quota binds. Total supply at seventy-five dollars is now too small, so consumers bid the price up. And that higher price pulls more domestic production in. In one sentence: by limiting imports, the government intends to raise the domestic price and domestic output, and so protect domestic producers — at the expense of consumers.

10 Q5.1 (v) tracing S + quota

0:55 · PRESENTER B

Part (v) is the part the question warns you about. The rule from Topic 5 is: when the domestic price is at or above the world price, supply becomes S plus quota. So the correct curve is kinked, not a straight line. Below seventy-five dollars there are no imports at all, so supply is just the domestic curve. At seventy-five dollars it jumps to the right by the quota — twenty thousand to forty thousand. Above that it is the domestic supply curve shifted right by twenty thousand, starting at forty thousand. The curve printed in the figure runs all the way down to twenty-five dollars on the price axis: it implies imports at prices where no imports are possible. That is the error they are testing.

11 Q5.1 (vi) quota rent

0:40 · PRESENTER B

Part (vi), quota rent. With S plus quota, the market clears where supply meets demand: eighty thousand televisions at a hundred and twenty-five dollars. Domestic producers supply sixty thousand of those, so imports are twenty thousand — exactly the quota, which is our check. Quota rent is the price gap times the quantity imported: a hundred and twenty-five minus seventy-five is fifty dollars, times twenty thousand — one million dollars. And notice who receives it: whoever holds the import licences, the importers, not the government. That is the difference from a tariff.

> Handover: Handover to [C] — the Q5.2 presenter takes over here.

12 Part 3 · Question 5.2

0:30 · PRESENTER C

Thank you — I will take Question 5.2. Country A is a small country again, and this time we are given equations: demand is minus two P plus forty thousand, supply is forty-six P plus six thousand four hundred. Parts (a) to (f): autarky, free-trade imports, the tariff that cuts imports by a quarter, tariff revenue, consumer surplus under the tariff, and finally whether a tariff benefits the country as a whole.

13 Q5.2 – setup

0:40 · PRESENTER C

Before the numbers, the method, because every part uses the same three moves. Without trade, the market clears domestically: set demand equal to supply. With trade, evaluate both equations at the world price of six hundred dollars. And imports are always demand minus supply at the price buyers actually face — which, with a tariff, is the world price plus the tariff. Subtract supply from demand once and you get the import function: thirty-three thousand six hundred minus forty-eight P. Every extra dollar of domestic price cuts imports by forty-eight handbags.

14 Q5.2 (a) autarky

0:30 · PRESENTER C

Part (a), autarky. Without international trade, quantity demanded equals quantity supplied. So minus two P plus forty thousand equals forty-six P plus six thousand four hundred. Collect the terms and thirty-three thousand six hundred equals forty-eight P, which gives an equilibrium price of seven hundred dollars. Put that back into demand and you get thirty-eight thousand six hundred handbags a year. Check it in supply — also thirty-eight thousand six hundred.

15 Q5.2 (b) free-trade imports

0:30 · PRESENTER C

Part (b), free trade at six hundred dollars a handbag. Demand at six hundred is thirty-eight thousand eight hundred; supply is thirty-four thousand. So imports are four thousand eight hundred handbags a year — or use the import function: thirty-three thousand six hundred minus forty-eight times six hundred gives the same answer. Consumers now pay less than they did under autarky, so they buy more; domestic firms sell less. That gap is the import.

16 Q5.2 (c) the tariff

0:45 · PRESENTER C

Part (c) is the interesting one. The government wants imports cut by twenty-five per cent, so the target is four thousand eight hundred times zero point seven five — three thousand six hundred handbags. What domestic price delivers that? Put three thousand six hundred into the import function: that gives forty-eight P equals thirty thousand, so P is six hundred and twenty-five dollars. The tariff is the gap between the price consumers pay and the world price: six hundred and twenty-five minus six hundred. Twenty-five dollars a handbag. Notice the logic runs backwards — the government chooses the import target, and the tariff is whatever delivers it.

17 Q5.2 (d) tariff revenue

0:20 · PRESENTER C

Part (d): tariff revenue is the tariff per unit multiplied by the number of units still imported — a rectangle, not a triangle. Twenty-five dollars on each of three thousand six hundred handbags is ninety thousand dollars a year. This is the number the statement in part (f) will point to, so remember it.

18 Q5.2 (e) consumer surplus

0:45 · PRESENTER C

Part (e), consumer surplus under the tariff. At six hundred and twenty-five dollars consumers buy thirty-eight thousand seven hundred and fifty handbags. The demand curve reaches the price axis where quantity demanded is zero — that is forty thousand divided by two, twenty thousand dollars. So the triangle is a half times thirty-eight thousand seven hundred and fifty times the height, twenty thousand minus six hundred and twenty-five. That gives three hundred and seventy-five million, three hundred and ninety thousand, six hundred and twenty-five dollars. Under free trade consumers had a million dollars more — the tariff costs them about nine hundred and sixty-nine thousand.

19 Q5.2 (f) is the tariff beneficial?

0:50 · PRESENTER C

Part (f), and this is the reflective one. The statement says that imposing a tariff is beneficial for country A as a whole because the government receives tariff revenue. It is false — for the country as a whole. Look at the three groups. Consumers lose nine hundred and sixty-nine thousand dollars. Domestic producers gain eight hundred and sixty-four thousand. The government collects ninety thousand. Add those up and the country is down fifteen thousand dollars — and that is exactly the deadweight loss: a half times the twenty-five dollar tariff times the twelve hundred handbags of imports that disappeared. So the statement is true for specific groups, the government and domestic producers, but the gains are transfers. A transfer cannot make the total any bigger.

> Handover: Handover to [D] — the Q5.3 and Q5.4 presenter takes over here.

20 Part 4 · Question 5.3

0:25 · PRESENTER D

I will finish with Questions 5.3 and 5.4. Question 5.3 is Egypt's generator market: demand ten thousand minus ten P, supply one thousand plus ten P, a world price of three hundred US dollars and a tariff of one hundred. Part (a) is arithmetic; part (b) asks who gains and who loses, and for the deadweight loss, illustrated with a diagram.

21 Q5.3 (a) Egypt

0:35 · PRESENTER D

Part (a). Once the tariff is in place, the price in Egypt is the world price plus the tariff: four hundred dollars. Domestic production at four hundred is one thousand plus ten times four hundred — five thousand generators. Domestic demand is ten thousand minus four thousand — six thousand. So imports are the difference: one thousand generators. Tariff revenue is a hundred dollars on each of them, one hundred thousand dollars. For comparison, under free trade Egypt imported three thousand — the tariff cut imports by two thirds.

22 Q5.3 (b) gainers and losers

0:50 · PRESENTER D

Part (b) — who gains and who loses. Producers gain: their surplus is the trapezoid between three hundred and four hundred dollars along the supply curve, and it is worth four hundred and fifty thousand. The government gains a hundred thousand in tariff revenue. Consumers lose: their surplus falls by the trapezoid along the demand curve, six hundred and fifty thousand. The two pieces nobody receives are the hatched triangles — the production distortion and the consumption distortion — fifty thousand each. So the country as a whole is down one hundred thousand dollars: minus six hundred and fifty, plus four hundred and fifty, plus one hundred. The same conclusion as the handbags: protection moves income around, and destroys some of it.

23 Part 5 · Question 5.4

0:20 · PRESENTER D

Question 5.4 asks us to choose one newspaper or magazine article from 2021 or later about a recent development in international trade, summarise it, and then apply Topic 5 to it. We chose Reuters' coverage of the European Union's tariffs on Chinese electric vehicles, published on the twenty-ninth of October 2024.

24 Q5.4 – EU tariffs on Chinese EVs

1:20 · PRESENTER D

What happened: after a year-long anti-subsidy investigation, the EU imposed definitive countervailing duties for five years, on top of the standard ten per cent. The rate depends on the company — BYD seventeen per cent, Geely eighteen point eight, SAIC thirty-five point three, Tesla's Shanghai plant seven point eight — so some models face forty-five per cent in total. EU members were divided, Beijing opened its own investigations into EU brandy, dairy and pork, and both sides kept negotiating a price undertaking instead. Now map it onto Topic 5. The barrier is an ad valorem tariff — a percentage tax on imports. EU consumers pay more, so their surplus falls. EU producers are protected from cheaper imports, so theirs rises. The EU budget collects the duty: tariff revenue. And fewer cars are imported, so the two deadweight-loss triangles appear. The price undertaking is the interesting line: if a minimum import price replaces the duty, the price gap stays with the Chinese exporters rather than the EU — that is the quota-rent case, exactly like a voluntary export restraint. And on comparative advantage: protection keeps production in a location where costs are higher.

25 Conclusion

0:40 · PRESENTER D

Three takeaways. First, both barriers raise the domestic price — a tariff fixes the price gap, a quota fixes the quantity and lets the price find its level. Second, someone always gets the gap: the government with a tariff, the licence holders with a quota — and if that rent goes abroad instead, the country is worse off still. And third: producers win, consumers lose, and the country as a whole loses by the two deadweight-loss triangles — fifteen thousand dollars in the handbag question, one hundred thousand in Egypt.

26 References and Q&A

0:15 · PRESENTER D

Our sources are on this slide: the Reuters article for Question 5.4, the tutorial question sheet, and Topic 5, Topic 2 and Unit 5 for the theory and the worked figures. Thank you — we are happy to take your questions.